Experiment (1)
Introduction to LT Spice

Objectives

- To simulate simple passive circuits using DC and Transient analysis.
- To simulate a first-order low-pass filter circuit using AC analysis.
- To be able to present results using plots for various circuit variables.

Pre-lab Work

Before starting this experiment, you need to study the document "Introduction to LT Spice IV with Examples", by Nicholas Lombardo, Linear Technology, 2015.

Circuits

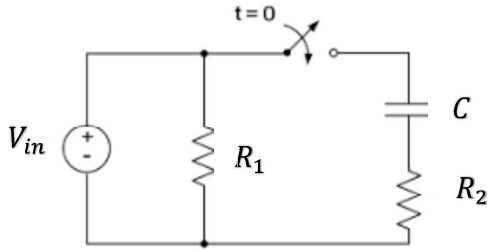


Figure 1: Transient circuit.

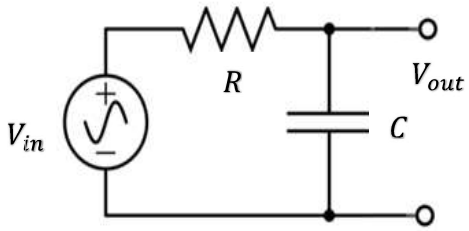


Figure 2: RC low-pass filter.

Procedure

Part (1): Transient Analysis.

1. Connect the circuit in figure 1. Use the circuit elements values given by the lab instructor. V_{in} is a DC voltage source.
2. With the switch open, simulate the circuit to find the current in R_1. Record your results in table 1.
3. Next, use the DC Sweep function and produce the IV curve for R_1.
4. Now with the switch closed, set V_{in} as a periodic rectangular signal with frequency f and maximum amplitude V_{max}.
5. Simulate the circuit using the Transient analysis function and generate the voltage waveform (versus time) across the capacitor C and the resistor R_2. Put these curves together with the V_{in} curve on the same plot.
6. Also observe the waveform of the current through R_2.
7. Change the value of f and repeat step 5.

Part (2): AC Analysis.

1. Connect the circuit in figure 2. Use the circuit elements values given by the lab instructor. V_{in} is a sinusoidal AC voltage source with frequency f and maximum amplitude V_{max}.
2. Simulate the circuit using the AC analysis function and generate the gain spectrum curve (gain versus frequency). Use dB scale for the gain and log scale for the frequency.
3. Change the value of C and observe the simulation curves.

Results and Analysis

Part (1): Transient Analysis.

Table 1. DC Simulations in Part (1).

DC Input Vin (V)	Simulation I_{R_1} (mA)	Calculation I_{R_1} (mA)	% Difference

1. Comment on the difference between simulation and theoretical values in table 1.

2. State your observations on the simulated behavior of the given RC circuit, including the effect of changing the input frequency.

Part (2): AC Analysis.

1. From your simulations, find the simulated bandwidth of the filter for different values of C. Record your results in table 2.

Table 2. Simulated and calculated bandwidth.

Capacitor value	Simulated Bandwidth	Theoretical Bandwidth	% Difference

2. Comment on the results in table 2.

3. If the output of the circuit is taken across the resistor instead of the capacitor, what kind of a filter is obtained? Explain your answer and support it using simulation results.

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